Lecture 24

Berry's phase

Objectives. *A slowly cycled Hamiltonian can return an eigenstate with a measurable relative phase. Part of this phase depends on the elapsed time. Another part depends on the path of the parameters. We identify the second part with line-bundle holonomy. For the aligned spin-1/2 ground state, it is minus half the signed solid angle enclosed by the field-direction loop. The eigenvalue gap states where this description applies.*

By the end of this lecture you will be able to:

- State the gap and slow-evolution hypotheses of the cited adiabatic theorem.
- Construct the connection and curvature of a smooth simple eigenline.
- Compute a closed-cycle geometric phase with the correct gauge convention.
- Derive the spin-1/2 phase and its ground-state Chern number -1.

24.1 Slow evolution and the gap

Recall the finite-dimensional quantum notation from *Quantum Mechanics I*. A pure state is a complex line in $\mathbb{C}^N$. A unit representative is written as a ket $|\psi\rangle$. Its conjugate transpose is the bra $\langle\psi|$, a linear functional. The inner product $\langle\phi|\psi\rangle = \phi^\dagger\psi$ is antilinear in its first argument. Multiplying a representative by a phase leaves its line unchanged. The Hamiltonian is a Hermitian matrix. Evolution satisfies $i\hbar\,\partial_t\,|\psi\rangle = H(t)\,|\psi\rangle$.

Let $s \in [0,1]$ describe a fixed path of external parameters. Executing it in a duration $T > 0$ means $s = t/T$ for $0 \le t \le T$. Write the evolving vector as $\psi_T(s)$. The *adiabatic limit* is $T \to \infty$ with this path fixed. An eigenvalue is *simple* if its eigenspace has dimension one. Its *gap* is its distance to the other eigenvalues.

Theorem 24.1 (Adiabatic evolution · *cited*)**.** *Let $H(s)$ be a smooth family of Hermitian $N \times N$ matrices on $[0,1]$. Let $E_n(s)$ be a smooth simple eigenvalue separated from all other eigenvalues by a fixed $\delta > 0$. Start in a unit vector of its eigenspace and solve $i\hbar\,\psi_T'(s) = TH(s)\psi_T(s)$. There is a unique smooth unit vector $v(s)$ in that eigenspace with $v(0) = \psi_T(0)$ and $\langle v(s)|v'(s)\rangle = 0$. For each fixed $s \in [0,1]$, the cited theorem gives*

$$\left\| \psi_T(s) - \exp\left(-\frac{iT}{\hbar}\int_0^s E_n(u)\,du\right) v(s) \right\| \longrightarrow 0 \quad \text{as } T \to \infty.$$

This is the finite-dimensional form of Kato's adiabatic theorem. See Barry Simon, p. 2167, for the gap assumption and phase separation, and p. 2168 for the transport condition on the limit. The exponential is the *dynamical phase factor*.

If $H(1) = H(0)$ and the chosen eigenline returns to itself, then $v(1) = e^{i\gamma(C)}v(0)$ for a phase defined modulo 2π. This scalar is the *geometric phase factor*, or *Berry phase factor*. It can be measured relative to a reference arm after accounting for the dynamical phase.

The theorem is a limit statement. A changing eigenvector need not solve the full Schrödinger equation exactly at finite T.

The figure separates a gap condition from the choice of eigenvectors.

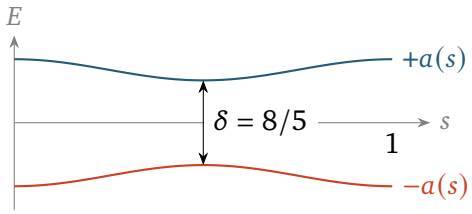


Figure 24.1: The diagonal family $H(s) = \mathrm{diag}(-a(s), a(s))$, with $a(s) = 1 + \cos(2\pi s)/5$, has a gap at least $8/5$. Its eigenspaces are constant, so its geometric phase is zero.

24.2 Eigenlines and their connection

Let $\mathcal{U}$ be an open region in a smooth parameter manifold, with a smooth family $H\colon \mathcal{U} \to \mathrm{Herm}(N)$ and a chosen smooth simple eigenvalue $E_n(R)$. Here $\mathrm{Herm}(N)$ is the real vector space of Hermitian matrices. The *eigenline bundle* is the subset of $\mathcal{U} \times \mathbb{C}^N$ given by

$$L_n = \{(R, w) : R \in \mathcal{U},\ H(R)w = E_n(R)w\}.$$

Its fibre over R is the eigenspace, including its zero vector. Smoothness of the selected branch is an explicit hypothesis here. It holds for the separated branches used below. Along a smooth parameter path $R(s)$, the adiabatic theorem applies to $H(R(s))$ when its eigenvalue has the stated uniform gap.

Theorem 24.2 (The Berry connection · *proved geometrically; adiabatic interpretation given the cited theorem*)**.** *The eigenlines form a smooth Hermitian line bundle. For a smooth unit eigenvector $|n_i\rangle$ on $U_i \subset \mathcal{U}$, the* Berry connection *has imaginary local one-form*

$$\mathcal{A}_i = \langle n_i | \mathrm{d} n_i \rangle\,.$$

These forms define the projected derivative $\nabla\xi = P\mathrm{d}\xi$, where $P(R)$ is the orthogonal projection onto the eigenline. Its parallel transport is the vector $v(s)$ in the adiabatic theorem. Thus its holonomy around a cycle C is $e^{\mathrm{i}\gamma(C)}$.

Proof. The matrix $H(R) - E_n(R)I$ has rank $N-1$. Its adjugate, the transposed cofactor matrix, has a nonzero column at any fixed point. Its columns lie in the kernel of the original matrix, by the identity $B\,\mathrm{adj}(B) = \det(B)I$ for every square matrix B. Choose such a column. It stays nonzero nearby and is smooth in R. Divide it by its positive norm to obtain a smooth unit eigenvector $|n_i\rangle$. Its scalar multiples give a local product chart for L_n. On overlaps write $|n_j\rangle = |n_i\rangle\, g_{ij}$, where $g_{ij} = \langle n_i | n_j \rangle \in \mathrm{U}(1)$. These smooth transitions satisfy the cocycle law, so Lecture 20's charts give the bundle.

The projection is $P = |n_i\rangle\langle n_i|$ and is unchanged by this unit phase. It is therefore global. For a section $\xi = |n_i\rangle\, c$, direct differentiation gives

$$P\mathrm{d}\xi = |n_i\rangle\,\big(\mathrm{d}c + \langle n_i|\mathrm{d}n_i\rangle\, c\big).$$

The ordinary product rule gives the connection Leibniz rule because $P\xi = \xi$. Self-adjointness of P similarly gives the Hermitian metric product rule. Differentiating $\langle n_i | n_i \rangle = 1$ shows

$\mathcal{A}_i + \overline{\mathcal{A}_i} = 0$, so the potential is imaginary. Finally, for a vector $v(s)$ in the eigenline, its projection of $v'(s)$ is $v(s)\,\langle v(s)|v'(s)\rangle$ when v is unit. Thus the adiabatic theorem's condition is exactly $\nabla_{\partial_s} v = 0$. Uniqueness of Lecture 21's transport identifies the two vectors and their endpoint phase. ∎

The bundle and connection depend only on the eigenspaces. The dynamical phase also uses the eigenvalues and traversal time.

24.3 Curvature, gauge changes and holonomy

A different unit vector on one chart has the form $|n'\rangle = |n\rangle\, h$, with $h\colon U \to \mathrm{U}(1)$. Differentiating the product and using unit norm gives

$$\mathcal{A}' = \mathcal{A} + h^{-1}\mathrm{d}h, \qquad \mathcal{A}' = \mathcal{A} + \mathrm{i}\mathrm{d}\chi \quad \text{if } h = e^{\mathrm{i}\chi}.$$

This is Lecture 21's passive line-frame law. The coefficient of a fixed section changes by $c' = h^{-1}c$. The *real Berry potential* is $A_{\mathrm{B}} = \mathrm{i}\mathcal{A}$. It differs from the imaginary connection form used in the transport equation.

The *Berry curvature* is the global two-form $\mathrm{d}\mathcal{A}$. Differentiating the local connection yields the full local formula

$$\mathrm{d}\mathcal{A} = \langle \mathrm{d}n| \wedge |\mathrm{d}n\rangle = \sum_{a=1}^{N} \mathrm{d}\overline{n^a} \wedge \mathrm{d}n^a.$$

Here the bra–ket wedge denotes the componentwise sum on the right. The term with $\mathrm{d}^2 n$ vanishes. The gauge law shows that the curvature is unchanged on overlaps. Thus Lecture 22 supplies the real class $c_1(L_n) = [\mathrm{i}\mathrm{d}\mathcal{A}/(2\pi)]$.

In a periodic unit frame along a closed cycle, the scalar transport equation gives

$$e^{\mathrm{i}\gamma(C)} = \exp\left(-\oint_C \mathcal{A}\right), \qquad \gamma(C) = \mathrm{i}\oint_C \mathcal{A} = \oint_C A_{\mathrm{B}} \pmod{2\pi}.$$

Such a frame exists along the cycle even when none exists on all of $\mathcal{U}$. Indeed, transport a unit vector along the parameter interval to get $v(s)$. If $v(1) = e^{\mathrm{i}\alpha} v(0)$, the vector $e^{-\mathrm{i}\alpha s} v(s)$ is periodic. Alternatively, use local frames and the transition factors when changing charts, as in Lecture 21. Under any single-valued gauge map on the cycle, $\oint_C h^{-1}\mathrm{d}h = 2\pi\mathrm{i}\,\mathrm{wind}(h)$. The geometric phase changes by an integer multiple of 2π, leaving its factor invariant. No global real logarithm of h is required.

If one frame covers an oriented compact spanning surface Σ with $\partial\Sigma = C$, Stokes gives $\gamma(C) = \mathrm{i}\int_\Sigma \mathrm{d}\mathcal{A}$ modulo 2π. The surface and frame hypotheses matter, as they did for the Aharonov–Bohm calculation. Berry's original equation (6) uses our $\mathrm{i}\oint \mathcal{A}$. Tong's current QHE notes instead use a real potential $A_{\mathrm{Tong}} = -\mathrm{i}\mathcal{A} = -A_{\mathrm{B}}$. His factor $\exp(-\mathrm{i}\oint A_{\mathrm{Tong}})$ is the same holonomy.

24.4 A spin-$1/2$ in a rotating field

Recall the Pauli matrices and the spin operators S_a, for $a \in \{x, y, z\}$:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad S_a = \frac{\hbar}{2}\sigma_a.$$

The symbol σ_a denotes these matrices, distinct from Lecture 21's section-coefficient column. Their products obey $\sigma_a\sigma_b = \delta_{ab}I + \mathrm{i}\epsilon_{abc}\sigma_c$. Use an energy-scaled field $\mathbf{B} = b\mathbf{n}$, with $b > 0$ and $|\mathbf{n}| = 1$. The Hamiltonian is $H = -\mathbf{B}\cdot\boldsymbol{\sigma}$. Its eigenvalues are $-b$ and $+b$ because $(\mathbf{n}\cdot\boldsymbol{\sigma})^2 = I$. Its gap is $2b$ everywhere on the direction sphere.

For a smooth loop $C\colon [0,1] \to S^2$ based at $p = C(0) = C(1)$, choose a continuous filling $f\colon [0,1]^2 \to S^2$, smooth on each rectangle of a finite subdivision, with

$$f(s,0) = C(s), \qquad f(s,1) = f(0,u) = f(1,u) = p.$$

Lecture 20's hemisphere, polygon and stereographic construction supplies such a filling, smooth on each rectangle of a finite subdivision. Orient the square by $\mathrm{d}s \wedge \mathrm{d}u$. Recall the outward unit-sphere area form β from Lecture 12. The *signed solid angle* of this filling is

$$\Omega_f(C) = \int_{[0,1]^2} f^*\beta.$$

The proof below shows that its class modulo 4π is independent of the filling. We denote this class by $\Omega(C)$.

Proposition 24.3 (The spin eigenline · *proved*)**.** *For* $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$*, the aligned ground-state eigenline is Lecture 20's tautological line on* $\mathbb{C}\mathrm{P}^1$*. On the northern chart it has*

$$|n_N\rangle = \begin{pmatrix} \cos(\theta/2) \\ e^{\mathrm{i}\phi}\sin(\theta/2) \end{pmatrix}, \quad \mathcal{A}_N = \frac{\mathrm{i}}{2}(1-\cos\theta)\mathrm{d}\phi, \quad \mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2}\sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

In the outward orientation its Chern number is -1*. For an oriented loop, the geometric phase is* $\gamma(C) = -\Omega(C)/2$ *modulo* 2π*, where the solid angle is defined modulo* 4π*.*

Proof. The direction matrix is

$$\mathbf{n}\cdot\boldsymbol{\sigma} = \begin{pmatrix} \cos\theta & e^{-\mathrm{i}\phi}\sin\theta \\ e^{\mathrm{i}\phi}\sin\theta & -\cos\theta \end{pmatrix}.$$

Multiplying by the displayed vector gives $|n_N\rangle$, so $H|n_N\rangle = -b|n_N\rangle$. Its norm is one. In $z = \tan(\theta/2)e^{\mathrm{i}\phi}$ it is $(1,z)^T/\sqrt{1+|z|^2}$, the tautological unit frame of Lecture 20. The inner product with its θ derivative is $-\cos(\theta/2)\sin(\theta/2)/2 + \sin(\theta/2)\cos(\theta/2)/2 = 0$. The ϕ derivative gives $\mathrm{i}\sin^2(\theta/2) = \mathrm{i}(1-\cos\theta)/2$. Differentiating gives the curvature stated above.

The southern unit vector is $|n_S\rangle = e^{-\mathrm{i}\phi}|n_N\rangle$. It is smooth at the south pole and has $\mathcal{A}_S = \mathcal{A}_N - \mathrm{i}\mathrm{d}\phi$. Thus its frame transition has winding -1. Equivalently, direct curvature integration gives

$$\frac{\mathrm{i}}{2\pi}\int_{S^2}\mathrm{d}\mathcal{A} = -\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi}\sin\theta\mathrm{d}\theta\mathrm{d}\phi = -1.$$

To compute the phase for a general loop, pull the eigenline and connection back to the filling square. Fix a unit vector over p. From the top edge, transport it down each vertical segment. Smooth dependence of transport on parameters gives a unit frame $w(s,u)$, continuous on the square and smooth on each rectangle of the finite subdivision. The top and side edges map constantly to p, so the frame is the same fixed vector on all three. In particular, its bottom restriction is a periodic frame along C.

Let $\mathcal{A}_f = \langle w|\mathrm{d}w\rangle$ be its connection form on the square. On each rectangle, its curvature is $\mathrm{d}\mathcal{A}_f = f^*\mathrm{d}\mathcal{A}$. The tangential restrictions on shared edges agree, so internal boundary integrals cancel. The top and side integrals vanish because their frame is constant. The

positively oriented bottom edge runs in increasing s. Stokes on the rectangles therefore gives

$$\gamma(C) = \mathrm{i}\int_0^1 \mathcal{A}_f(\partial_s)(s,0)\mathrm{d}s = \mathrm{i}\int_{[0,1]^2} f^*\mathrm{d}\mathcal{A} = -\frac{1}{2}\Omega_f(C) \pmod{2\pi}.$$

Two fillings f and g compute the same intrinsic holonomy. Hence $\exp[-\mathrm{i}(\Omega_f(C)-\Omega_g(C))/2] = 1$, so their signed areas differ by an integer multiple of 4π. This proves both the solid-angle convention and the claimed phase formula. ∎

The figure fixes the orientation of a latitude and its northern cap.

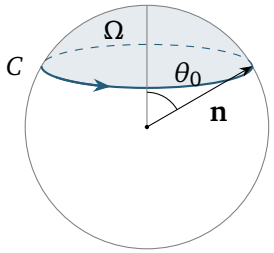


Figure 24.2: Increasing azimuth orients the latitude as the boundary of the northern cap. The cap has solid angle $\Omega = 2\pi(1-\cos\theta_0)$. Its spin ground-state geometric phase is $-\Omega/2$.

Example 24.4 (A closed field rotation)**.** Let $b > 0$ be fixed, choose $0 < \theta_0 < \pi$, and rotate the field with $\phi(t) = 2\pi t/T$ for $0 \le t \le T$. Start in $|n_N(\theta_0, 0)\rangle$ and take the adiabatic limit. The northern unit frame is periodic on this path, so

$$\gamma(C) = \mathrm{i}\int_0^{2\pi} \frac{\mathrm{i}}{2}(1-\cos\theta_0)\mathrm{d}\phi = -\pi(1-\cos\theta_0).$$

The energy is $-b$, so the dynamical factor is $e^{+\mathrm{i}bT/\hbar}$. The final state is asymptotic to

$$e^{+\mathrm{i}bT/\hbar}e^{-\mathrm{i}\pi(1-\cos\theta_0)}\,|n_N(\theta_0,0)\rangle\,.$$

For $\theta_0 = \pi/3$ the geometric factor is $-\mathrm{i}$. The southern potential instead gives a phase larger by 2π, hence the same factor. Using the same cap, Lecture 18's tangent-vector transport angle is Ω. The spin phase is its negative half before either angle is reduced modulo 2π. At the equator the vector holonomy is a full turn while the spin phase factor is -1. The orthogonal excited vector $(-e^{-\mathrm{i}\phi}\sin(\theta/2), \cos(\theta/2))^T$ has $\mathcal{A}_{\mathrm{exc}} = -\mathrm{i}(1-\cos\theta)\mathrm{d}\phi/2$. Its curvature and Chern number have the opposite sign, giving $+1$.

24.5 The degeneracy and general spin

For $\mathbf{B} \neq 0$, the ground-state projector is $P(\mathbf{B}) = (I + \mathbf{B}\cdot\boldsymbol{\sigma}/|\mathbf{B}|)/2$. It depends only on the field direction. Pulling the sphere curvature back to $\mathbb{R}^3 \setminus \{0\}$ gives

$$\mathrm{d}\mathcal{A} = \frac{\mathrm{i}}{2|\mathbf{B}|^3}\left(B^1\mathrm{d}B^2\wedge\mathrm{d}B^3 + B^2\mathrm{d}B^3\wedge\mathrm{d}B^1 + B^3\mathrm{d}B^1\wedge\mathrm{d}B^2\right).$$

The real curvature $\mathrm{d}A_{\mathrm{B}} = \mathrm{i}\mathrm{d}\mathcal{A}$ points inward. Its flux through an outward-oriented sphere is -2π. The figure shows this parameter-space monopole.

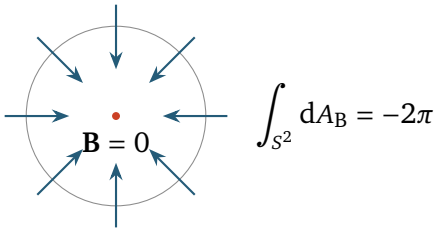


Figure 24.3: The real Berry curvature points inward in field-parameter space. Any outward sphere around the excluded degeneracy has $\int dA_B = -2\pi$ and ground-line Chern number -1.

At $\mathbf{B} = 0$ the two eigenvalues coincide. Every line is then an eigenline, so the selected ground line and its projector are not determined. For the identically zero Hamiltonian the exact evolution is constant. It does not force a prescribed changing eigenline to be followed. The gap hypothesis prevents this ambiguity.

Proposition 24.5 (Spin j · *sketched*)**.** *Let $j > 0$ be an integer or half-integer. Write the dimensionless spin matrices as $J_a = S_a/\hbar$, with $[J_a, J_b] = i\epsilon_{abc}J_c$. On the outward direction sphere, for the eigenline of $\mathbf{n}\cdot\mathbf{J}$ of weight $m \in \{-j, -j+1, \ldots, j\}$,*

$$\mathcal{A}_N = im(1-\cos\theta)d\phi, \qquad \gamma(C) = -m\Omega(C) \pmod{2\pi}, \qquad C_1 = -2m.$$

Sketch. Choose $J_z|j,m\rangle = m|j,m\rangle$ and rotate the weight vector by

$$|n_N\rangle = e^{im\phi}e^{-i\phi J_z}e^{-i\theta J_y}|j,m\rangle .$$

The phase makes this vector single-valued in ϕ, since $m - j \in \mathbb{Z}$. The angular-momentum commutators give $e^{i\theta J_y}J_z e^{-i\theta J_y} = J_z\cos\theta - J_x\sin\theta$. In a weight vector, $\langle j,m|J_x|j,m\rangle = \langle j,m|J_y|j,m\rangle = 0$ and $\langle j,m|J_z|j,m\rangle = m$. Thus the θ coefficient is zero and the ϕ coefficient is $im - im\cos\theta$. Differentiate, integrate over the sphere, and apply the transport formula to get the stated values. The rotation and weight identities are the standard angular-momentum input recalled from *Quantum Mechanics I*. Berry, §4, pp. 50–51, obtains the same spin phase using ladder operators. ∎

The quantity $2m$ is an integer, so changing the solid angle by 4π does not change the phase factor. For the ground line of $H = -\kappa\mathbf{B}\cdot\mathbf{J}$ with $\kappa > 0$, the weight is $m = j$ and the degeneracy is again at $\mathbf{B} = 0$.

Self-check. For a smooth open path $C\colon [0,1] \to \mathcal{U}$, explain how $i\int_C \mathcal{A}$ changes under $|n\rangle \mapsto e^{i\chi}|n\rangle$, for a single-valued real χ on the path.

Answer. It changes by $-[\chi(C(1)) - \chi(C(0))]$. Thus this integral alone is not an intrinsic open-path phase. An endpoint reference must also transform if an open-path interference measurement is specified. For a closed cycle the holonomy is intrinsic and its phase is defined modulo 2π.

Summary. A smooth simple eigenline has the unitary connection $\mathcal{A} = \langle n|\mathrm{d}n\rangle$ and curvature $\langle \mathrm{d}n| \wedge |\mathrm{d}n\rangle$. The cited gapped adiabatic theorem identifies its holonomy with the geometric factor after removing the dynamical phase. The imaginary connection and real potential satisfy $A_{\mathrm{B}} = \mathrm{i}\mathcal{A}$. For $H = -\mathbf{B}\cdot\boldsymbol{\sigma}$ with $\mathbf{B} \neq 0$, the ground line has Chern number -1 on an outward direction sphere and $\gamma = -\Omega/2$ modulo 2π. The excluded degeneracy is the source of this parameter-space curvature. Lecture 25 integrates the same curvature over the momentum torus of a gapped band.

Reading. Berry, *Quantal phase factors accompanying adiabatic changes*, §2 and §4, pp. 46–48 and 50–51. Simon, *Holonomy, the quantum adiabatic theorem and Berry's phase*, pp. 2167–2168. Nakahara, §10.6. Tong, *The Quantum Hall Effect*, current §1.5.1–1.5.2, pp. 28–34, with $A_{\mathrm{Tong}} = -A_{\mathrm{B}}$.